MD. TAWFIQUL ISLAM TAMAL

Computer Science & Engineering Student | Data Science & Machine Learning



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Professional Summary

I am a Computer Science and Engineering student at American International University-Bangladesh (AIUB), specializing in Data Science and Machine Learning. As an applied ML researcher, I bring a strong record of academic publication, including three conference-presented papers, one accepted paper, and a self-authored machine learning reference book, alongside hands-on expertise in explainable ensemble modeling, SHAP interpretability, and statistical analysis. I combine rigorous research capability with proven leadership experience to translate complex data into deployable, real-world solutions across healthcare, sustainability, and finance.

Education

American International University-Bangladesh (AIUB) — 2023 – 2027 (Ongoing)
BSc in Computer Science & Engineering, Major in Information Systems — Dhaka, Bangladesh
CGPA: 3.55/4.00

Rajendrapur Cantonment Public School and College (RCPSC) — 2020 – 2022
Higher Secondary Certificate (HSC), Science Group — Bangladesh
GPA: 5.00/5.00

Bangladesh Rice Research Institute (BRRI) High School — 2019 – 2020
Secondary School Certificate (SSC), Science Group — Bangladesh
GPA: 4.72/5.00

Technical Skills

- **Languages:** Python, SQL (MySQL), C++, C#, Java, JavaScript, PHP, R, Assembly, HTML
- **Machine Learning & Data Science:** Scikit-learn, SHAP, Ensemble Methods (Stacking, XGBoost, Random Forest, SVM, AdaBoost, PSO-optimization), Statistical Analysis, Data Analysis & Visualization
- **Tools & Platforms:** Git/GitHub, Cisco Packet Tracer, Microsoft Sentinel, Power BI, LaTeX, MS Excel/Word/PowerPoint
- **Operating Systems:** Linux (Ubuntu), Windows, macOS
- **Core Concepts:** Explainable AI (XAI), Predictive Modeling, Research & Academic Writing

Experience

Campus Ambassador — Sep 2026 – Present
Jaatra – Dhaka, Mapped Smarter — Dhaka, Bangladesh (Remote)

- Represented Jaatra at the university level, building organizational presence among students and promoting the vision of smarter mobility.
- Participated in ambassador sessions and community meetings, supporting campaigns and engaging peers to drive awareness.
- Shared feedback and contributed ideas that helped shape product features, collaborating with the team on projects and community development.

Campus Director Aug 2026 – Present
Elevetr AI Dhaka, Bangladesh (Remote)

- Represented Elevetr AI at the university level, building organizational presence and driving AI-awareness initiatives on campus.
- Participated in orientation and leadership sessions, engaging students and supporting outreach programs.
- Collaborated with the Elevetr AI team to foster innovation and create professional development opportunities for students.

Campus Ambassador Aug 2026 – Present
STechNest Dhaka, Bangladesh (Remote)

- Represented STechNest at the university level, building organizational presence among students.
- Participated in ambassador sessions and supported campaigns to promote innovation and growth among peers.

Senior Research Ambassador Aug 2026 – Present
Education Reform Bangladesh – Youth Research Program 2026 (YRP 2026) Dhaka, Bangladesh (Remote)

- Contributed to a nationwide research initiative on Higher Education and Employability in Bangladesh, in collaboration with SRIC and Sunaru Academy.
- Supported evidence-based research, collecting and analyzing data to identify policy and skills gaps.

Freelancer 2021 – Present
Upwork Remote

- Delivered data-driven digital projects and content solutions for international clients, managing requirements, timelines, and communication throughout the project lifecycle.

Projects

Cardiovascular Risk Prediction with Explainable Machine Learning Code
Tech Stack: Python, Scikit-learn, XGBoost, LightGBM, SHAP

- Developed a heart disease risk prediction model comparing Logistic Regression, Random Forest, XGBoost, and LightGBM.
- Applied SHAP-based explainability and a fairness check across demographic subgroups.

Amazon Product Reviews Analytics Code
Tech Stack: Python, NLP, Web Scraping

- Built an end-to-end analytics pipeline (web scraping, EDA, NLP sentiment analysis) processing 34,000+ real product reviews.
- Achieved 87.4% agreement with star ratings.

FaceSenseAIPro Code
Tech Stack: Python, Computer Vision

- Built a facial recognition and analysis application applying real-time computer vision techniques.

Foodly – Online Food Blog Code
Tech Stack: PHP 8, MySQL, JavaScript

- Engineered a PHP MVC web application for restaurant/food browsing, member reviews, and admin moderation.

Smart Gas Leak Detection System Code
Tech Stack: Sensors, Embedded Systems

- Developed a sensor-based safety system to detect hazardous gas leaks and trigger real-time alerts.

Traffic Signal Control System Code
Tech Stack: Simulation, Signal-Timing Logic

- Designed a simulated automated traffic control system modeling real-world signal-timing logic.

Lost in Dimensions: Journey Back to Earth Code
Tech Stack: C++, OpenGL

- Co-developed a 2D game using C++ and OpenGL graphics programming.

Uthao – A Ride-Sharing Management System Code
Tech Stack: System Design, Role-Based Access Control

- Designed a ride-sharing management system for helicopter services, featuring flight booking, real-time GPS tracking, and role-based access control.

Personal Utilities & Web Apps Code
Tech Stack: HTML, CSS, JavaScript

- Built a collection of interactive web applications (Quiz System, Dice Roller, Calculator, Digital Calendar) to strengthen front-end development skills.

Publications & Research

- AI-Driven Smart Waste Management System for Sustainable Urban Development in Bangladesh — 8th International Conference on Sustainable Development (ICSD 2026), United International University (UIU), April 2026 (Presented — not yet published).
- Data-Driven Analysis of Textile Waste Generation and Recycling Potential in Bangladesh — 4th International Conference on Textile Science and Engineering (ICTSE 2026), Bangladesh University of Textiles (BUTEX), Dhaka, June 2026 (Presented — not yet published).
- Machine Learning: From Zero to Advanced — A Comprehensive Guide to Concepts, Models, Mathematics, Visualization, and Applied Research — self-authored technical book, July 2026 (ResearchGate).
- Credit Card Default Prediction Using Machine Learning and Statistical Analysis Techniques — 4th International Conference on Computing Advancements (ICCA 2026), American International University-Bangladesh (AIUB) (Accepted).
- SmartWasteAI: An AI-Based Framework for Feasibility and Public Acceptance Assessment in Sustainable Urban Waste Management for Bangladesh — 29th International Conference on Computer and Information Technology (ICCIT 2026), September 2026 (Submitted).
- BnCodeSent: A Bag-of-Words and TF-IDF Based Machine Learning Framework for Bangla-English Code-Mixed Sentiment Classification — 6th International Conference on Trends in Electronics and Health Informatics (TEHI-2026), University of Creative Technology Chittagong (UCTC), September 2026 (Submitted).
- An Explainable Ensemble Machine Learning Framework for Flood Risk Prediction Using Hydrological and Meteorological Data — 14th International Conference on Electrical and Computer Engineering (ICECE 2026), Bangladesh University of Engineering and Technology (BUET), August 2026 (Submitted).

Ongoing Research

- ShrimpSentinelNet: An Attention-Based Deep Convolutional Neural Network with Stacked Ensemble Learning and Multi-Method Explainable AI for Automated Shrimp Disease Classification in Aquaculture — manuscript in preparation, targeted for journal submission.
- An Explainable Deep Learning Framework for EEG-Based Brain Signal Classification in BCI Systems — manuscript in preparation, targeted for journal submission.
- Quantitative Assessment of Textile Waste Dynamics and Recycling Readiness: Evidence-Based Sustainability Insights for the Apparel Sector — manuscript in preparation, targeted for journal submission.
- Predicting Environmental Degradation from Globalization and Energy Dynamics: A Machine Learning Classification Framework for Global Sustainability Assessment — manuscript in preparation, targeted for journal submission.

Certifications & Training

- IEEE SPAVe 7.0 & 8.0 — Participant, IEEE AIUB Student Branch (2024, 2025)
- Fundamentals of Research & Basic Data Analysis — Research Gateway, 2-month internship (Dec 2025 – Jan 2026)
- Threat Detection with Microsoft Sentinel Analytics — Microsoft (Mar 2026)
- Explain Threat Hunting Concepts in Microsoft Sentinel — Microsoft (Mar 2026)
- AI for Recruitment: Resume Screening System from Scratch — Kryzotech (Mar 2026)
- Fundamentals of Qualitative Research Methodology — Neeti Gobeshona Kendro (May 2026)
- Business Analytics with Power BI — Unbolt Academy (Jun 2026)
- Introduction to Medical Image Analysis using Python for Health Research — BIIHR (Jun 2026)
- STATA, SPSS, R, Python, AI for Research & Higher Studies — Betterskill.io (Jul 2026)
- AI-Powered Analytics — Grameenphone Academy (Aug 2026)
- ML Bootcamp by Shohoj Skill & Shohoj Coding — IEEE CIS IUBAT SBC (Aug 2026)
- AI-Driven Software Engineering: Shaping the Future of Development — IEEE CS Premier University Student Branch Chapter (Sep 2026)
- Data Analytics Essentials — Cisco Networking Academy (2026)
- Getting Started with Cisco Packet Tracer — Cisco Networking Academy (2026)
- Statistical Data Analysis and Visualization with R Programming

Languages

Bangla — Native (Advanced) — English — Intermediate — German — Basic

References

Prof. Dr. Dip Nandi — Professor & Associate Dean, Faculty of Science & Technology, AIUB
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